



Dr Dominika Malinowska

Remote Sensing and ML Specialist

[linkedin.com/in/malinowskadinika](https://www.linkedin.com/in/malinowskadinika)



github.com/dominika-malinowska



dominika.u.malinowska@gmail.com



AI/ML engineer and researcher with 6+ years' experience in Python for geospatial analytics. Expertise in converting complex SAR and optical data into actionable monitoring tools and automated pipelines to improve data usability for decision-making, backed by high-impact publications. Skilled at leading interdisciplinary collaborations and communicating technical concepts to partners such as ESA and industry stakeholders. Eager to deploy robust ML models and operationalise multi-sensor Earth Observation workflows to deliver critical, scalable monitoring solutions.

Professional Experience

R&D Machine Learning Expert | Spottitt (now part of Lucey Group), Poland/Remote

06/2025 - Present

- Led and contributed to data science and AI projects across the group, including the development of a commercial SAR-based product for house subsidence monitoring
- Achieved a ~75% reduction in false positives for a key power line monitoring project by integrating an effective off-the-shelf cloud masking solution into the production pipeline
- Engineered an ML pipeline for object height estimation from singular optical images, using PyTorch/TorchGeo to adapt research in neural network architectures for commercial tasks

Doctoral Researcher | University of Bath, UK/Remote

12/2019 – 11/2025

Visiting Researcher | TU Delft, NL/Remote

03/2020 – 11/2025

- Led a global geo-hazard risk assessment of long-span bridges using InSAR availability with a team of 5, proving Sentinel-1 operational viability for worldwide infrastructure monitoring. Project resulted in a first-author publication in Nature Communications (selected for Editors' Highlights, DOI: [10.1038/s41467-025-64260-x](https://doi.org/10.1038/s41467-025-64260-x))
- Built a U-Net-based model to predict Multi-Temporal Interferometric Synthetic Aperture Radar (MT-InSAR) measurement points density, improving Sentinel-1 data usability for infrastructure monitoring; published findings in IEEE JSTARS (DOI: [10.1109/JSTARS.2024.3449688](https://doi.org/10.1109/JSTARS.2024.3449688))
- Developed a multidisciplinary framework to prioritise SAR-based monitoring for improving safety of vulnerable communities affected by natural hazards and ensure socially equitable use of the data for bridge maintenance

Researcher | Frontier Development Lab (FDL) Europe, Remote

06/2024 – 10/2024

- Co-led an 8-person R&D project to develop an ML pipeline for solar flare prediction for the ESA Vigil 2.0 mission, managing task planning, workload distribution, and leading meetings with academic and industry partners
- Collaborated with the team to propose innovative labelling methods, deliver technical reports with results to ESA stakeholders within tight deadlines, and present findings via [live stream](#)

Intern in Earth Observation (EO) Group | SCISYS Space Division (now part of CGI), UK

07/2019 – 08/2019

- Contributed to software optimisation by evaluating autonomous detection of features on optical imagery, identifying key performance issues for further improvement
- Supported EO project proposals development, aligning early-stage research with tender requirements

Software Test Engineer | SCISYS Space Division (now part of CGI), UK

04/2018 – 09/2018

- Defined and executed manual tests, reported anomalies, and verified requirements with a traceability matrix to ensure comprehensive test coverage for satellite constellations scheduling software.

Education

PhD in Civil Engineering (specialising in Remote Sensing) | University of Bath, UK

12/2019 – 11/2025

- Advanced the application of MT-InSAR for infrastructure monitoring and hazard mitigation by proposing novel methodologies integrating ML, structural analysis, risk assessment, and social vulnerability

MSc in Astronautics and Space Engineering | Cranfield University, UK

09/2018 – 09/2019

- Conducted a system-level analysis for integrating a CubeSat laboratory (BAMMSat) into flight opportunities. Graduated with Distinction and received Course Director's Prize for outstanding performance

Leadership & Selected Projects

MSc students' supervision:

- Co-supervised two postgraduate projects, supporting the processing of Sentinel-1 data for heritage site (presentation at [EGU24](#)) and glacial lakes monitoring and co-authoring two journal papers

European Space Agency (ESA) PhiWeek Bootcamp 2019 Solving Big Energy Challenges with EO:

- Contributed to a wildfire detection system prototype for powerline protection by collaborating with a multidisciplinary team, exploring user needs, and presenting the final solution to potential clients

BAMMsat on ESA Balloon Experiments for University Students (BEXUS):

- Led a team of 15 MSc students as a System Engineer for a CubeSat-sized laboratory on a high-altitude balloon, facilitating team collaboration, co-writing the proposal, and presenting to a panel of experts at ESTEC

Laser Interferometer Space Antenna (LISA) Communication Relay Mission:

- Analysed the feasibility of a ground-spacecraft communication subsystem as part of an MSc group design project, presented results and recommendations to professionals at Airbus Defence & Space

Technical Competencies

Remote Sensing & Image Processing:

- **Data Types:** proficient in processing both optical and radar (SAR) datasets
- **Data Standards:** worked with GeoTIFF, GeoJSON, and KML
- **Software:** QGIS, Google Earth Engine (GEE), Google Earth Pro
- **Analysis:** using EO to address user requirements with geospatial analysis

Scientific Programming & Data Analysis:

- **Python Stack:** advanced proficiency in PyTorch, TorchGeo, Rasterio, GDAL, NumPy, GeoPandas, and Lightning
- **Machine Learning:** regression, Principal Component Analysis (PCA), and advanced ML algorithm implementation
- **Cloud & HPC:** experience in code deployment on Google Cloud Platform (GCP), AWS, and High-Performance Computing systems, as well as managing cloud storage via S3 and GCP buckets
- **Optimisation:** expertise in parallel programming (Mpi4py) and code optimisation for scientific datasets and large-scale analysis

Technical Environments:

- **OS:** Linux and Windows
- **Tools:** Git, shell scripting, VS Code, and Cursor

Behavioural Competencies

- **Communication:** skilled in translating complex technical concepts to diverse audiences via technical reports, manuals, and live presentations
- **Collaboration:** experienced in leading interdisciplinary teams and practising Non-Violent Communication to foster inclusive environments
- **Mentorship:** proven ability to supervise and mentor technical teams, ensuring complex geospatial workflows are understood and adopted.
- **Management:** proven ability to manage complex tasks and meet tight deadlines while working independently across multiple organisations and institutions
- **Problem-solving:** demonstrated ability to handle technical challenges and deliver results under pressure in fast-paced environments
- **Adaptability:** fast learner of new technical domains and tools, with a proactive approach to continuous professional improvement
- **Inclusion:** committed to inclusive environments, with training in Unconscious Bias and a focus on equitable data use

Language Skills

Polish: Native proficiency

English: Professional proficiency